

Drumil Patel

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PROFESSIONAL EXPERIENCE

Senior Member of Technical Staff

07/2024 – Present | Bangalore, India

Devrev

- Owned the architecture for DevRev's native observability warehouse (MELTS - Metrics, Events, Logs, Traces & Sessions), powering observability for enterprise customers including Indian Banks(HDFC, ICICI, Axis Bank, Bajaj, etc.), as well as internal use cases spanning agent evaluation, commerce, and incident management

Query Engine

- Designed and built a distributed SQL query engine using Apache DataFusion and Ballista, integrating Tantivy for full-text search, Bloom filter-based indexes for trace search, and Parquet-based columnar storage on S3, enabling highly parallel, low-latency query execution at PB scale.
- Architected custom materialized view integration in DataFusion's logical plan conversion, enabling transparent querying across raw and precomputed data for faster analytics under the same unified SQL interface.
- Built an extensible UDF framework enabling FDEs to integrate domain-specific logic into the warehouse, with anomaly detection and log pattern mining (Drain3) as example use cases.

Ingestion

- Designed a horizontally scalable ingestion pipeline sustaining **35 TB/hour of compressed telemetry and 5–6 PB of monthly volume**, replacing a legacy MongoDB/EBS architecture with a self-healing, elastic S3-based storage layer that decouples compute from storage and enables independent scaling of ingestion and persistence.
- Achieved 5× higher ingestion capacity while reducing infrastructure costs from \$100K → \$17K/month (~\$1M annually) through compute/storage decoupling and elastic, usage-based S3 storage; extended the architecture with compliance-grade **BYOB and BYOK** primitives for regulated enterprise customers.
- Architected MELTS' downstream ecosystem extensibility layer, enabling a range of use cases built directly on the MELTS warehouse including commerce, incident management workflows, LLM/Harness observability and agent-improvement pipelines
- Tech Stack Used:** Rust, Golang, DataFusion Ballista, Fluentd, Tantivy, Parquet, S3

Platform Engineer

05/2022 – 06/2024 | Bangalore, India

Newton School, Incanus Technologies

- Developed control plane for learning environments similar to replit, Github Codespaces which can spawn interactable environments(pods) for students like Jupyter Notebooks, Nodejs + Express. **Tech Stack Used:** Go, K8s SDK, Crossplane.
- Designed an event-driven infrastructure with cutting edge for ML and AI production workloads for Kubernetes, seamlessly integrating DAPR and Kafka as universal event brokers. This architecture serves as the foundation for successfully deploying over 15+ models into production. **Tech Stack Used:** Kafka, DAPR, Flask, Unicorn, Kubernetes

Software Engineering Intern

06/2021 – 07/2021 | Hyderabad, India

Amazon

- Build a centralized configuration store for maintaining uniformity of configurations across all FBA Export Services using Amazon's internal Tree based database, with custom java library for extraction of configuration based on service locations.
- Tech Stack Used:** Java, Spring boot, Amazon Web Services, G2S2(Amazon Internal Tree based data structure)

Prometheus, Google Summer of Code

05/2020 – 08/2020 | Remote

Cloud Native Computing Foundation

- Added EKS-cluster support to Prombench, Prometheus's Kubernetes-based benchmarking control plane, orchestrating ~1,000 pods across scrapers and readers to stress-test Prometheus performance at scale (GSoC blog [🔗](#))
- Tech Stack Used:** Golang, Kubernetes, Prometheus, AWS, GCP

Submittly, Google Summer of Code

05/2019 – 08/2019 | Remote

Rensselaer Polytechnic Institute, New York

- Implemented randomized input/output generation to prevent hardcoded solutions and ensure authentic assessment, while engineering a RabbitMQ-backed, worker-driven evaluation infrastructure capable of processing **10,000 students per hour** and **60,000 submissions during exam windows**. **Tech Stack Used:** Python, C++, RabbitMQ, PostgreSQL, Symphony

EDUCATION

Indian Institute of Technology, Roorkee

06/2018 – 06/2022 | Roorkee, India

Bachelor of Technology, Electrical Engineering

PUBLICATIONS

Random Input and Automated Output Generation in Submittly [🔗](#)

ACM Technical Symposium on Computing Science Education (SIGCSE)

2020